

# Changing shape

All animals change size and shape as they grow up. In some animals, however, the young and adults look completely different and live in different ways. The transformation from young to adult often happens gradually in a process called metamorphosis—for example, when a tadpole turns into a newt or a frog. But for other animals, including butterflies, metamorphosis is a single dramatic event. The animal's entire body breaks down into a soup, and a new body forms in its place.

## Complete transformation

Like all butterflies, the painted lady undergoes an amazing transformation called complete metamorphosis. It starts its life as a crawling caterpillar, which feeds hungrily on leaves for about four weeks. Next, the caterpillar attaches itself to a stem and spins a silk case called a chrysalis. Inside the chrysalis, the caterpillar's body is first liquefied, and then the cells reorganize themselves in the new shape of a butterfly. Finally, the chrysalis splits open and the adult butterfly emerges with crumpled wings.



Caterpillar hangs upside down to spin its chrysalis



Caterpillar's body changes inside the hanging chrysalis



## Right side up

Flatfish, such as this flounder, start life swimming in open water with an eye on each side of their heads, just like other fish. As the fish grow, their bodies flatten out and they lie on one side, partially buried on the seabed. Over time, the eye that was facing downward gradually creeps across the fish's head, forming a mismatched pair on the upper side of its body.

*Chrysalis shell remains in place after butterfly climbs out*



Adult painted lady pulls itself free of the chrysalis case

*Wings expand and harden as blood pumps through them*